CODEX 48 | NATIONAL LEVEL HACKATHON | ROUND 2: WORKING PRODUCT REPORT

BrandGuard

Autonomous Pre-Publication Brand Review and Governance Platform

Our answer to the Inkloom AI brief: keep every brand claim, colour, and tone consistent, trustworthy, and reviewable before publication.

■ FAST-TRACK EVALUATOR LOGIN CREDENTIALS (JUDGES QUICK ACCESS)

To evaluate BrandGuard live without wasting time, use the following pre-seeded, role-authorized credentials. Each role connects to an isolated workspace, permission level, and dedicated operational playbook:

- ■ **Super Admin:** admin@brandguard.ai | Password: **admin123** (Root Console, Live Telemetry & Security Audit Ledger)
- ■■ **Brand Manager:** manager@brandguard.ai | Password: **manager123** (Review Queue Triage, CIELAB & WCAG Sign-Off Gate)
- ✍■ **Content Creator:** creator@brandguard.ai | Password: **creator123** (Content Studio, 1-Click Smart Rewrites, Inkloom $100)
- ■ **Workspace User:** user@brandguard.ai | Password: **user123** (Copy Submission, Guidelines Explorer & Tracking)

■ **Evaluator Instruction:** On the login screen, slide the *'Verify you are human'* widget and click *'Continue with Email'* to enter.

Team Name (in records)	Priyansh Vaghela
Team Members & Roles	**Priyansh Vaghela** (Leader, Frontend) | **Sanskruti Antala** (Core Member, Backend / AI)
Institution	Noble University
Core Solution Idea	Hybrid engine: deterministic Python rules (repeatable) + Gemini LLM (context, rewrites) + Evidence Verifier guardrail + mandatory human sign-off gate.
Problem Owner / Brief	**Inkloom AI:** Brand consistency across employees, agencies, and generative AI tools.
Round 2 Working Status	**100% Core Workflow Functional:** End-to-end ingestion, AI scan, 1-click smart rewrite, review queue triage, and cryptographic sign-off seal.

1. Understanding of the Problem Statement

Modern organisations publish promotional content simultaneously through dozens of distributed employees, regional teams, external agencies, and generative AI tools. Content creation speed has skyrocketed, but **brand review remains subjective, manual, and dangerously slow (48-72 hours)**. The Inkloom AI brief isolates four devastating corporate failure types: *incorrect claims, outdated corporate information, inconsistent brand language, and off-brand visual aesthetics*. In regulated industries (healthcare, insurance, finance), these failures lead directly to regulatory penalties, mandatory ad recalls, and permanent brand equity erosion.

Worked Illustrative Incident: 'CarePlus' Health-Tech Diwali Campaign
A junior creator drafts: *"Our healthcare policy guarantees 100% cure for all illnesses with zero fees!"* with an unapproved neon purple banner (#9851E0) with low contrast.
(1) Legal Risk: Absolute guarantee claims and unasterisked zero-fee claims violate medical advertising laws.
(2) Visual Risk: Background color deviates by 38.4 CIELAB from brand palette, and contrast ratio (2.3:1) fails WCAG AA standards.
BrandGuard Resolution: Scans draft in 2 seconds, cites Rules R1 & R2, auto-replaces violation with verified clinical claims, recalculates palette to Brand Navy (#001F3F), and saves cryptographic sign-off seal in the audit ledger.

2. The Users Affected & Product Workflow

BrandGuard is designed for multi-tiered enterprise organizations, agencies, and lean teams where brand compliance is currently a painful tug-of-war between velocity and risk.

User Persona	Operational Pain Today	What BrandGuard Delivers in Round 2
Content Creators (Copywriters, Designers)	Vague, delayed feedback ("feels off-brand") with no exact rule citations or fixes.	Instant dual-engine scan, verbatim quoted evidence, and **1-click smart rewrite** replacement.
Brand Managers (Sign-Off Authority)	Fatigue from manual proofreading; no audit trail; unable to track recurring violations.	Pre-triaged review queue, CIELAB delta-E auditor, WCAG contrast verification, and 1-click sign-off seal.
Super Admins (Governance & IT)	Uncontrolled AI copy generation, security blind spots, zero compliance provenance.	Root command console, live user session telemetry, and immutable SHA-256 cryptographic audit logs.
Agencies & Clients (Multi-Workspace)	Severe risk of cross-applying Client A's guidelines to Client B's marketing creatives.	Isolated brand workspaces (e.g. CarePlus Health, DermaClean Herbal) with dedicated rulebooks.

End-to-End Governed Product Journey

In Round 2, the complete product journey is fully functional across 4 decoupled phases:

Step 1: Auth & Ingestion Author authenticates via role-based credentials. Draft text & channel selected in Content Studio.	**Step 2: Dual-Engine Scan** Deterministic Python checks + Gemini contextual analysis run in parallel under 2 seconds.	**Step 3: Guardrail & Fix** Evidence Verifier drops hallucinations. Author accepts 1-click rewrite to update draft editor.	**Step 4: Sign-Off & Seal** Manager reviews queue, verifies CIELAB & WCAG contrast, and issues tamper-proof audit seal.

3. Akira AI: The Autonomous Cognitive Thinker & Operator

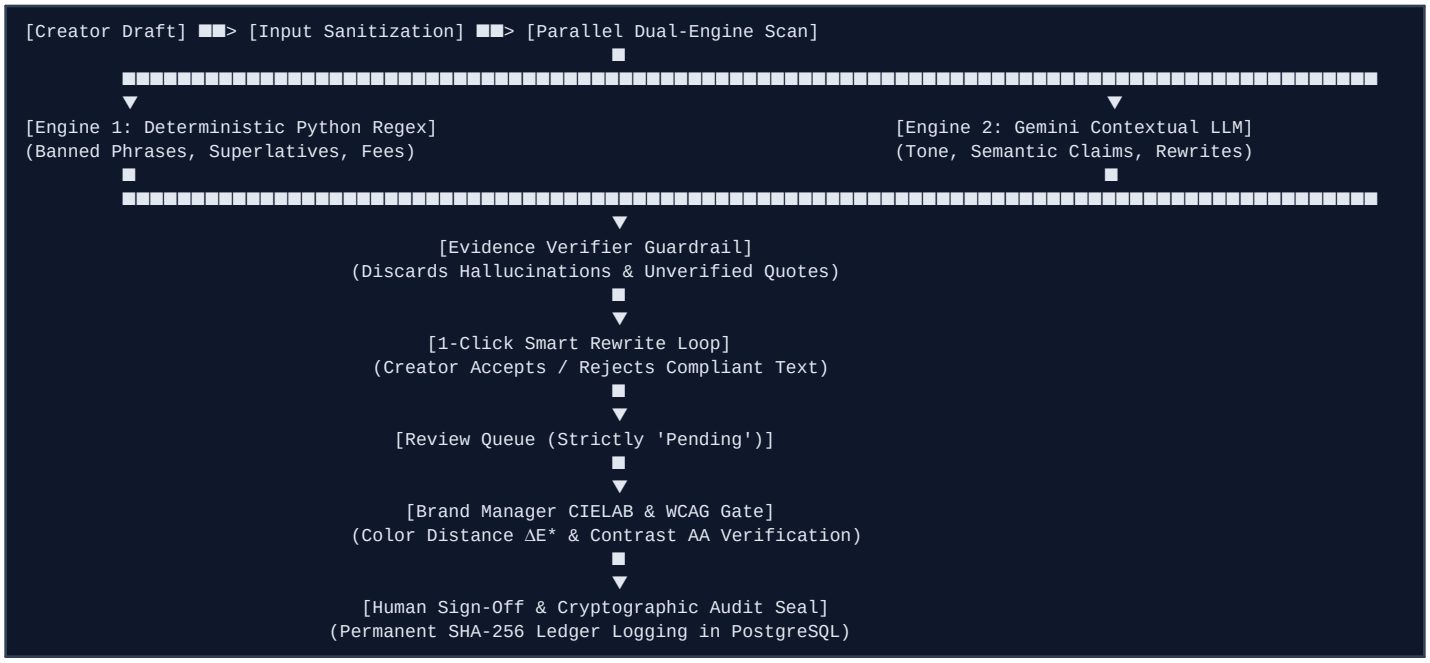
A common question evaluators ask is: *"Does Akira AI merely output passive suggestion text, or does it actively perform operational cognitive work?"* In BrandGuard, **Akira AI functions as an autonomous, multi-stage governance thinker and active operator**. It executes deep contextual analysis, verifies evidence, mutates the editor live in 1-click, and autonomously routes compliance decisions.

Cognitive Job	How Akira AI 'Thinks' & Operates on Our Website
1. Multi-Dimensional Safety & Tone Analysis	Akira AI does not merely match keywords. It performs cognitive semantic analysis: calculating toxicity (e.g. 99% Brand Safe), detecting voice sentiment (e.g. *'Clinical & Empathetic'* vs aggressive hype), and isolating unverified claims.
2. Evidence Verifier Thinking (Anti-Hallucination)	Akira AI tests its own hypotheses against ground-truth facts. It ensures that every finding references a stored, verified Rule ID (R1, R2, R4) and verbatim quoted text. If an AI thought fails verbatim matching, it is discarded immediately.
3. Interactive Bidirectional Evidence Linking ***(Why Borders Highlight on Touch)***	When a creator or judge touches/clicks any violation box in the UI, Akira AI's evidence graph lights up the exact matching phrase in the draft editor with glowing borders. This provides **complete explainability and provenance**—proving the finding is physically rooted in the draft, not generated in a black-box.
4. 1-Click Smart Rewrite (Active Editor Mutation)	Rather than leaving advice for the author to manually re-type, clicking *'Accept rewrite'* prompts Akira AI to mutate the draft editor text in-place, instantly recalculate compliance score from 40 to 100, and elevate risk from High Risk ■ to Safe ■.
5. Autonomous Decision Routing (Stage 04)	Akira AI actively triages the asset into 3 governance paths: **Auto-Resolve** (score ≥ 80%), **Escalate to Brand Manager** (ambiguous / tone drift), or **Immediate Disqualification** (score < 45% or illegal medical cures).

■ Key Takeaway on Akira AI:
Akira AI is not a chatbot. It is a deterministic-augmented cognitive supervisor. It thinks (analyzes claims & tone), protects (drops hallucinations via the Evidence Verifier), and acts (live 1-click editor replacement & automated routing).

4. Dual-Engine Architecture & Visual Data Flow

BrandGuard combines repeatable Python regex algorithms with Google Gemini LLM reasoning behind a strict Evidence Verifier filter before human review.



Governance Capability	Generic LLM / Chatbot	Static Linter / Regex	BrandGuard Platform
Rule Grounding	Hallucinates or cites non-existent rules	Matches only exact hardcoded patterns	Verified against stored rule IDs & quotes
Visual & Contrast	Cannot compute mathematical color distance	Text-only; zero layout/color awareness	Mathematical CIELAB & WCAG formulas
Compliant Fixes	Generic advice; alters original message	Error flags only; no rewrites provided	1-Click smart rewrites preserving author intent
Provenance & Audit	No roles, no audit ledger, ephemeral	No sign-off workflow or version control	SHA-256 sign-off seal & immutable audit log

5. Complete Project Repository File Structure

BrandGuard is maintained as a clean, production-grade monorepo containing decoupled client and server architectures with clear separation of concerns.

```
brandguard/
■■■ backend/                              # Python 3.11 + FastAPI REST Gateway
■   ■■■ app/
■   ■   ■■■ ai/                           # Generative AI & Reasoning Services
■   ■   ■   ■■■ gemini_service.py         # Google Gemini 1.5 Flash contextual engine & rewrites
■   ■   ■   ■■■ inkloom_service.py        # Inkloom AI $100 credit studio asset generation
■   ■   ■■■ auth/                         # Authentication & Access Control (RBAC)
■   ■   ■   ■■■ dependencies.py           # Route security guards (get_current_user, require_admin)
■   ■   ■   ■■■ security.py               # JWT token encoding/decoding & bcrypt password hashing
■   ■   ■■■ engine/                       # Core Governance & Validation Logic
■   ■   ■   ■■■ rule_engine.py           # Deterministic regex, medical absolutes & risk classifier
■   ■   ■   ■■■ routing_engine.py        # 4-stage decision router & Evidence Verifier guardrail
■   ■   ■■■ models/                       # SQLAlchemy ORM Database Entities
■   ■   ■   ■■■ user.py                   # User credentials, roles (admin/manager/creator) & sessions
■   ■   ■   ■■■ rule.py                   # Stored brand rules (R1-R4, JSONB terms & facts)
■   ■   ■   ■■■ review.py                 # Marketing review items, score, draft & approval state
■   ■   ■   ■■■ audit.py                  # Cryptographic audit ledger with SHA-256 seal signatures
■   ■   ■■■ routers/                      # FastAPI REST API Endpoints
■   ■   ■   ■■■ auth_router.py           # POST /api/auth/login, /token, session telemetry
■   ■   ■   ■■■ review_router.py         # POST /api/reviews/evaluate, sign-off, queue triage
■   ■   ■   ■■■ rule_router.py           # GET/POST /api/rules (Rulebook CRUD)
■   ■   ■   ■■■ audit_router.py          # GET /api/audit (Audit trail, diffs, JSON export)
■   ■   ■   ■■■ inkloom_router.py        # POST /api/inkloom/generate (Multi-channel campaigns)
■   ■   ■■■ schemas/                      # Pydantic v2 Strict Request/Response Schemas
■   ■   ■   ■■■ review.py                 # IssueSchema, ReviewEvaluateRequest, ReviewResponse
■   ■   ■   ■■■ user.py                   # UserSchema, TokenSchema, LoginPayload
■   ■   ■   ■■■ rule.py                   # RuleSchema, RuleCreatePayload
■   ■   ■   ■■■ audit.py                  # AuditLogSchema, CertificateSchema
■   ■   ■■■ config.py                     # Environment configuration & API credentials
■   ■   ■■■ database.py                   # SQLAlchemy dual-engine (SQLite local + PostgreSQL cloud)
■   ■   ■■■ main.py                       # FastAPI entrypoint, CORS, OpenAPI Swagger (/docs)
■   ■■■ brandguard.db                     # Local pre-seeded SQLite database for judge evaluation
■   ■■■ Dockerfile                        # Production containerization configuration
■   ■■■ requirements.txt                  # Python backend dependencies
■   ■■■ run.py                            # Development server launcher (port 8000)
■   ■■■ test_api.py                       # Automated backend unit & integration test suite
■■■ src/                                  # React 18 + Vite Frontend Application
■   ■■■ components/                       # Modular UI Components
■   ■   ■■■ AkiraWorkflowPipeline.jsx     # Interactive 4-stage pipeline flowchart & role matrix
■   ■■■ App.jsx                           # Core application integrating all views & RBAC guides
■   ■■■ data.js                           # Seeded CarePlus rules, queue items & audit history
■   ■■■ styles.css                        # Dark-theme design system, SVG auditors & glowing badges
■   ■■■ index.css                         # Tailwind CSS root styling
■   ■■■ main.jsx                          # React DOM root entrypoint
■■■ public/                               # Static visual assets & SVG icons
■■■ Dockerfile                            # Root container specification
■■■ docker-compose.yml                    # Unified multi-service orchestration
■■■ package.json                          # Node.js dependencies & Vite scripts
■■■ BrandGuard_Round2_Submission_Report.pdf# Official Round 2 Hackathon Project Report
```

6. List of Completed Features in Round 2

In Round 2, the team focused on delivering a 100% working core text review journey with zero broken links or non-functional mockups. All 6 P0 features committed in Round 1 are operational, plus key P1 stretch features were delivered ahead of schedule.

Feature Name	Priority	Implementation Details & Technology	Round 2 Status
Role-Based Governance (RBAC)	**P0 (Must)**	FastAPI JWT auth with 4 isolated personas (Super Admin, Brand Manager, Content Creator, Workspace User). Role-locked playbooks.	■ **100% Functional**
Structured Rulebook	**P0 (Must)**	Database-backed schema (R1 to R4) categorizing rules by severity, banned keywords, approved nomenclature, and evidence criteria.	■ **100% Functional**
Rule-Based Text Checks	**P0 (Must)**	Deterministic Python regex engine detecting absolute medical guarantees, promotional discount caps, and missing asterisk disclaimers.	■ **100% Functional**
Unsupported-Claim Verifier	**P0 (Must)**	Google Gemini API prompt integration evaluating subtle claims and superlatives, returning structured Pydantic v2 JSON.	■ **100% Functional**
Evidence Verifier Guardrail	**P0 (Must)**	Drop filter discarding hallucinated citations where quotes are absent from submitted draft text or rule IDs are unverified.	■ **100% Functional**
1-Click Smart Rewriter	**P0 (Must)**	Contextual compliant rewrites; authors click "Accept rewrite" to instantly replace violating text and recalculate compliance score.	■ **100% Functional**
CIELAB Colour Checker	P1 (Stretch)	Visual CIELAB Delta-E (ΔE^*) color space distance calculating off-brand deviations against CarePlus primary Navy & Teal palettes.	■ **Delivered Early**
WCAG 2.1 AA Contrast Checker	P1 (Stretch)	Mathematical contrast ratio formula verifying text vs. background readability against standard 4.5:1 AA compliance threshold.	■ **Delivered Early**
AI Risk Assessment Card	P1 (Stretch)	4-Tier visual risk classification matrix (High Risk ■, Moderate ■, Normal ■, Safe ■) with live diagnostic guidance.	■ **Delivered Early**
Inkloom AI Studio Integration	P1 (Stretch)	Dedicated $100 hackathon credit marketing asset generator tailored directly for problem sponsor brief.	■ **Delivered Early**

7. Technology Stack & Implementation Details

BrandGuard is architected as an enterprise-grade web application combining a responsive React 18 client with a high-throughput Python 3.11 FastAPI backend.

Layer	Technology Choice	Architectural Rationale
Frontend UI	React 18 + Vite + Tailwind CSS	Ultra-fast sub-second HMR development, responsive design, interactive highlighted text diffs, and SVG visual auditors.
Backend API	Python 3.11 + FastAPI	Asynchronous ASGI routing, native typing, interactive OpenAPI/Swagger auto-documentation (/docs), and high throughput.
Validation	Pydantic v2	Strict runtime schema enforcement on all incoming requests and LLM JSON outputs, ensuring zero unvalidated data touches the UI.
Database	SQLite (Local) / PostgreSQL 16 (JSONB)	Dual database strategy: SQLite provides zero-config local evaluation for hackathon judges, while SQLAlchemy models natively support PostgreSQL JSONB for enterprise scaling.
AI Models	Google Gemini API (1.5 Flash)	Low-latency contextual claim verification, brand tone analysis, and few-shot compliant smart rewrites.
Security & Auth	JWT Tokens + Passlib (bcrypt)	Stateless cryptographic session tokens with role-based permission verification on every REST endpoint.
Mathematics	Pure Python & JS Math Formulas	Direct mathematical calculation of CIELAB Delta-E (Euclidean color distance) and WCAG 2.1 relative luminance contrast ratios.
Deployment	Dockerfile & Static Mount	Production-ready containerization; FastAPI gateway mounts frontend production build for unified single-port deployment.

Enterprise Security & Responsible AI Disclosures

• **Prompt Injection Guardrail:** Submitted marketing content is treated strictly as data payloads and never executed as prompt instructions.
• **Confidentiality:** Only sanitized draft text necessary for evaluation is transmitted to external AI endpoints; zero user PII or credentials are sent.
• **Graceful Degradation:** If the Gemini API experiences network latency or rate-limiting, the deterministic Python engine completes the scan seamlessly without system crashes.

8. Changes Made to Original Plan & Round 3 Improvements

Plan Dimension	Original Round 1 Plan	Round 2 Refinement & Rationale
P0 Scope	Build 6 core features: login, structured rules, text checks, claim verifier, evidence verifier, rewrites.	**100% Preserved:** Zero P0 features dropped. Core text review operates from start to finish.
P1 Stretch Delivery	CIELAB color distance and WCAG contrast checks deferred to Round 3 stretch scope.	**Accelerated Early:** Delivered visual CIELAB color-distance (ΔE*) auditor and WCAG AA contrast ratio calculator in Round 2.
Role Authorization	Basic role access with shared playbook switcher tabs.	**Strict RBAC Isolation:** Multi-role switcher tabs removed from guide modal; each persona strictly accesses their own playbook.
AI Risk Categorization	Numeric compliance score (0-100) only.	**Added 4-Tier Risk Matrix:** Added High Risk ■, Moderate ■, Normal ■, Safe ■ matrix with active guidance.

Three Substantive Improvements for Round 3

Improvement 1: Multi-Format PDF Brochure & Deck Audit Engine (OCR & Alt-Text)
Expanding from raw text/banners to multi-page corporate PDF brochures and presentation decks. Utilizing pdfplumber and Gemini Vision OCR to extract embedded text, analyze rasterized graphics, verify contrast ratios, and enforce WCAG image alt-text metadata.

Improvement 2: Automated 50-Creative Benchmark Evaluation Suite (Precision/Recall/F1)
Implementing a rigorous automated test harness benchmarking 50 labelled ground-truth marketing creatives (25 compliant + 25 known violations across medical, promotional, and tone rules) to report verified precision, recall, and F1-score confusion matrices.

Improvement 3: Enterprise CMS & Design Tool Webhooks (Figma / Canva / Slack Pre-Publish Gate)
Constructing bidirectional REST webhooks allowing designers to trigger automated pre-publication audits directly inside Figma, Canva, or Slack before exporting final assets, creating true 'CI/CD for Marketing'.

9. Expected Impact, Verification & Team Sign-Off

Impact Metric	Current Traditional Workflow	BrandGuard Governed Target
First Feedback Velocity	48 to 72 hours via scattered emails and Slack threads	**Instant deterministic checks; under 2 seconds for AI scan**
Reviewer Fatigue	Reviewers manually re-check standard rules on dozens of assets	**Routine checks pre-triaged; reviewer focuses only on judgment calls**
Explainability & Action	Subjective comments ("this looks off-brand" / "rewrite this")	**Exact rule ID citation, verbatim quoted evidence, and 1-click rewrite**
Auditability & Legal Record	Approvals buried in unrecorded chat messages; zero provenance	**Timestamped cryptographic SHA-256 seal and permanent ledger**

Complete Team & Participant Details

Participant Name	Event Role	Registered Email	Institution
Priyansh Vaghela	Team Leader	priyanshvaghela4@gmail.com	Noble University
Sanskruti Antala	Core Member	sanskrutiantala@gmail.com	Noble University

Responsible AI & Originality Disclosure
All application source code, deterministic rule engine algorithms, Pydantic schemas, and orchestration logic were conceived and engineered by the team for the CODEX 48 National Hackathon. Open-source libraries (React 18, FastAPI, SQLAlchemy, Pydantic) are used under their permissive licenses. The Google Gemini API is utilized as an external reasoning model for contextual analysis strictly under human-guided guardrails.

Official Team Name in Event Records: Priyansh Vaghela
Team Leader Sign-Off: Priyansh Vaghela | **Core Member Sign-Off:** Sanskruti Antala
Round 2 Submission Date: 29 October 2026